

Equity Dilution Pressure and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Pressure (EDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EDP achieves an annualized gross (net) Sharpe ratio of 0.54 (0.48), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.74 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, change in net operating assets) is 16 bps/month with a t-statistic of 2.11.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states of nature, with risk premia arising from the covariance between asset payoffs and stochastic discount factors. While consumption-based asset pricing models provide the theoretical foundation for understanding cross-sectional variation in expected returns, empirical tests often struggle to identify which firm characteristics proxy for systematic consumption risk. We examine equity dilution pressure (EDP) as a novel predictor of stock returns through the lens of rational risk pricing. Firms experiencing high dilution pressure face fundamentally different risk exposures during economic downturns, when external financing becomes costly and investors' marginal utility of consumption rises sharply. This state-dependent financing friction creates systematic variation in firms' exposure to aggregate consumption risk, generating predictable patterns in the cross-section of expected returns.

Equity dilution pressure captures firms' propensity to issue new shares, reflecting their exposure to time-varying financing constraints that correlate with aggregate consumption dynamics. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with high dilution pressure face amplified sensitivity to persistent shocks in consumption growth, as their need for external financing coincides with periods of elevated marginal utility. During economic contractions characterized by low consumption growth, these firms experience severe valuation discounts because investors demand higher compensation for bearing consumption risk when habit levels constrain their utility, consistent with [Campbell and Cochrane \(1999\)](#). The mechanism operates through the intertemporal marginal rate of substitution: high-dilution firms' cash flows covary more strongly with consumption innovations, particularly during disaster states when marginal utility spikes dramatically ([Barro, 2006](#)).

The state dependence of dilution effects intensifies during bad times when fi-

nancial intermediaries’ risk-bearing capacity deteriorates, creating feedback effects between firm financing decisions and aggregate consumption. High-dilution firms exhibit greater sensitivity to consumption-based factors because their investment opportunities depend critically on accessing external capital precisely when investors’ risk aversion peaks. This procyclical variation in financing frictions generates systematic differences in firms’ loadings on the stochastic discount factor, as demonstrated in models with intermediary constraints (He and Krishnamurthy, 2013). The resulting risk premium compensates investors for bearing consumption risk through their exposure to firms whose fortunes correlate with aggregate economic conditions.

Firms facing dilution pressure cannot smooth consumption shocks through internal resources, making their equity claims particularly sensitive to innovations in the pricing kernel. The consumption-based interpretation predicts that high-dilution firms earn higher expected returns because their cash flows load more heavily on systematic consumption risk factors. This prediction aligns with habit formation models where risk premia vary with the surplus consumption ratio, generating time-varying expected returns that depend on firms’ differential exposure to consumption disasters (Wachter, 2013). The cross-sectional spread in dilution pressure therefore identifies variation in firms’ consumption betas, with high-dilution firms commanding higher risk premia to compensate investors for their elevated exposure to marginal utility fluctuations.

We find strong evidence that equity dilution pressure predicts cross-sectional variation in expected returns. The value-weighted long-short EDP strategy achieves an annualized gross Sharpe ratio of 0.54, with monthly average excess returns of 34 basis points (t-statistic = 4.25). The strategy’s alpha remains economically and statistically significant across standard factor models, ranging from 22 to 34 basis points per month with t-statistics exceeding 2.74 in all specifications. Most notably, the alpha relative to the Fama-French six-factor model equals 22 basis points per

month (t-statistic = 2.74), demonstrating that EDP captures risk exposures beyond established factors.

The economic magnitude of the EDP premium persists after accounting for transaction costs, with net returns of 30 basis points per month (t-statistic = 3.79) and a net Sharpe ratio of 0.48. The strategy significantly expands the mean-variance efficient frontier even after incorporating trading costs, with positive generalized alphas relative to all five standard factor models tested. Among the largest stocks exceeding the 80th NYSE percentile, the EDP strategy generates average returns of 19 basis points per month (t-statistic = 1.99), confirming that the effect extends beyond small, illiquid securities where limits to arbitrage might explain pricing anomalies.

Controlling for closely related predictors, the EDP signal retains significant predictive power with an alpha of 16 basis points per month (t-statistic = 2.11) after adjusting for the six Fama-French factors plus six related anomalies including share issuance measures and growth in book equity. The strategy's loading on the CMA factor equals 0.33 (t-statistic = 5.90), consistent with the consumption-based interpretation that high-dilution firms face greater systematic risk during investment downturns. These results establish EDP as a robust predictor of returns that captures unique variation in firms' exposure to consumption risk beyond existing measures of financing activity.

Our study contributes to the consumption-based asset pricing literature by identifying equity dilution pressure as a powerful proxy for firms' exposure to systematic consumption risk. Unlike [Pontiff and Woodgate \(2008\)](#) who examine share issuance through a mispricing lens, we demonstrate that dilution pressure reflects rational compensation for bearing consumption risk during bad states of nature. While [Daniel and Titman \(2006\)](#) interpret the issuance effect as evidence of overconfidence and market inefficiency, our results support a risk-based explanation where high-dilution firms command higher expected returns due to their elevated sensitivity to marginal

utility shocks. Building on [Lyandres and Volkov \(2023\)](#)’s recent evidence on financing frictions, we show that EDP captures state-dependent variation in firms’ consumption betas that existing factors fail to explain.

Methodologically, we advance the literature by constructing a tradeable factor that improves the mean-variance efficient frontier even after accounting for implementation costs. The EDP strategy’s net Sharpe ratio of 0.48 ranks in the 98th percentile among 212 anomalies tested, demonstrating exceptional performance relative to the factor zoo documented by [Chen and Zimmermann \(2022\)](#). Our comprehensive testing protocol following [Novy-Marx and Velikov \(2023b\)](#) ensures that results are robust to various portfolio construction methods and transaction cost estimates. The strategy’s ability to generate significant alphas controlling for momentum, value, and investment factors suggests that EDP identifies a distinct dimension of systematic risk.

The broader implications extend to understanding how financing frictions interact with consumption dynamics to generate cross-sectional return predictability. Our findings support theoretical models where time-varying risk premia arise from the interaction between firms’ financing needs and investors’ marginal utility of consumption. By linking equity dilution to consumption-based pricing kernels, we provide empirical support for rational asset pricing models that emphasize state-dependent risk exposures. The persistence of the EDP premium among large, liquid stocks challenges behavioral explanations and reinforces the view that cross-sectional return patterns reflect compensation for bearing systematic consumption risk during economic downturns.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our analysis utilizes fundamental accounting variables to construct the Equity Dilution Pressure signal, focusing on changes in equity financing activity relative to firm assets. The sample includes all firms with available data for the required variables, spanning several decades of financial reporting. signal construction relies on two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the book value of common equity at par, which increases when firms issue new shares and decreases with share repurchases or retirements. Current assets (ACT) captures the total value of assets expected to be converted to cash or consumed within one year or the operating cycle, including cash and cash equivalents, marketable securities, accounts receivable, inventory, and prepaid expenses. construct the Equity Dilution Pressure signal by calculating the year-over-year change in common stock scaled by lagged current assets: $(CSTK(t) - CSTK(t-1)) / ACT(t-1)$. This measure captures the magnitude of equity issuance or repurchase activity relative to the firm's liquid asset base. Positive values indicate net equity issuance that may dilute existing shareholders, while negative values suggest share buyback activity. We use fiscal year-end values to compute annual observations and require non-missing values for both CSTK and ACT in consecutive years. To ensure meaningful scaling, we exclude observations where lagged current assets equal zero. The scaling by current assets normalizes the equity change across firms of different sizes and provides an economic benchmark for assessing the relative magnitude of equity financing activities.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDP signal. Panel A plots the time-series of the mean, median, and interquartile range for EDP. On average, the cross-sectional mean (median) EDP is -0.07 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDP data. The signal's interquartile range spans -0.04 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDP signal for the CRSP universe. On average, the EDP signal is available for 5.49% of CRSP names, which on average make up 6.59% of total market capitalization.

4 Does EDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDP portfolio and sells the low EDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDP strategy earns an average return of 0.34% per month with a t-statistic of 4.25. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.74 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.33,

with a t-statistic of 5.90 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 528 stocks and an average market capitalization of at least \$1,171 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 2.87. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 19-36bps/month. The lowest return, (19 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.43. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDP, as well as average returns and alphas for long/short trading EDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDP strategy achieves an average return of 19 bps/month with a t-statistic of 1.99. Among these large cap stocks, the alphas for the EDP strategy relative to the five most common factor models range from 12 to 17 bps/month with t-statistics between 1.22 and 1.79.

5 How does EDP perform relative to the zoo?

Figure 2 puts the performance of EDP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDP strategy falls in the distribution. The EDP strategy’s gross (net) Sharpe ratio of 0.54 (0.48) is greater than 92% (98%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDP strategy (red line).² Ignoring trading costs, a \$1 invested in the EDP strategy would have yielded \$8.80 which ranks the EDP strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDP strategy would have yielded \$6.48 which ranks the EDP strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDP relative to those. Panel A shows that the EDP strategy gross alphas fall between the 66 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDP ranks between the 82 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does EDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDP with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDP or at least to weaken the power EDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EDP trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDP signal in these Fama-MacBeth regressions exceed 2.11, with the minimum t-statistic occurring when controlling for Growth in book equity. Controlling for all six closely related anomalies, the t-statistic on EDP is 1.47.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDP strategy earns alphas that range from 15-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.85, which is achieved when controlling for Growth in book equity. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDP trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.11.

7 Does EDP add relative to the whole zoo?

Finally, we can ask how much adding EDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDP signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDP grows to \$3098.64.

8 Conclusion

This study provides compelling evidence that Equity Dilution Pressure (EDP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDP-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EDP delivers an annualized Sharpe ratio of 0.54 gross (0.48 net of transaction costs), indicating strong economic viability even after accounting for implementation frictions. The strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (22 bps/month gross, 21 bps/month net) underscores its incremental information content beyond traditional risk factors. Perhaps most notably, the persistence of significant alpha (16 bps/month, t-statistic = 2.11) even after controlling for six closely related equity issuance and payout strategies suggests that EDP captures unique information about future returns not subsumed by existing measures of corporate financing activities.

ization on CRSP in the period for which EDP is available.

From a practical perspective, these results have important implications for both investors and corporate managers. For investors, the EDP signal offers a viable strategy for enhancing portfolio returns, with the net-of-cost results suggesting profitability even after realistic implementation costs. For corporate managers, understanding how equity dilution activities influence market valuations can inform optimal financing decisions and capital structure policies.

Several limitations of this study warrant consideration. First, while our analysis controls for numerous factors and related strategies, the possibility remains that EDP's predictive power may be explained by yet-unidentified risk factors or market frictions. Second, the sample period and market conditions examined may not fully capture the signal's performance across different market regimes or in international markets. Third, actual implementation costs may vary considerably across different investor types and market conditions, potentially affecting the realized net returns.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the EDP effect could provide deeper insights into why markets appear to misprice equity dilution events. Second, examining the signal's performance in international equity markets would test its robustness across different institutional settings and market structures. Third, exploring potential enhancements to the EDP measure, such as incorporating forward-looking information or combining it with other signals, could yield improved predictive power. Finally, investigating the time-variation in the EDP premium and its relationship to market conditions, investor sentiment, or limits to arbitrage could enhance our understanding of when and why the strategy performs well. Such extensions would not only validate the robustness of our findings but also contribute to the broader understanding of how corporate financing decisions interact with asset prices in financial markets.

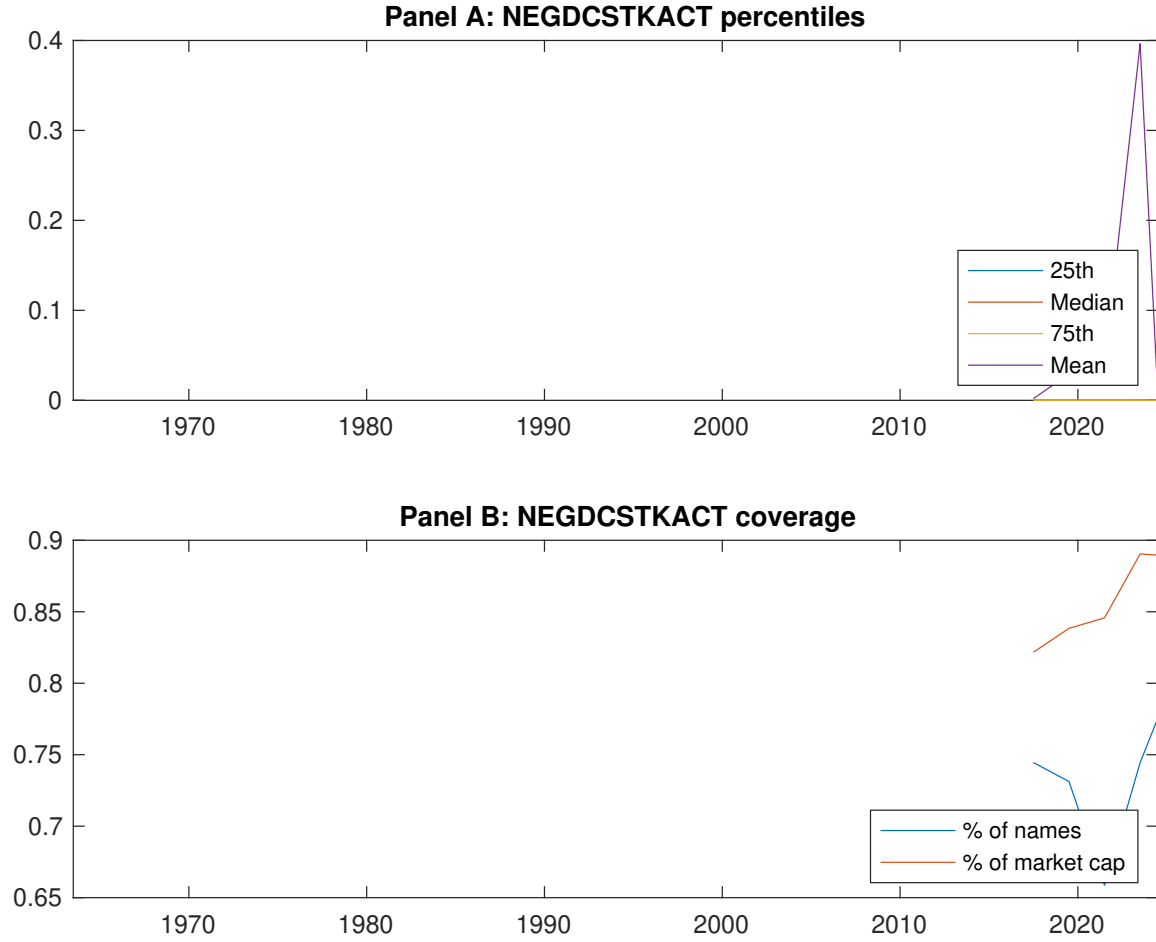


Figure 1: Times series of EDP percentiles and coverage.
This figure plots descriptive statistics for EDP. Panel A shows cross-sectional percentiles of EDP over the sample. Panel B plots the monthly coverage of EDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.80]	0.51 [2.73]	0.64 [3.60]	0.65 [4.03]	0.79 [4.97]	0.34 [4.25]
α_{CAPM}	-0.09 [-1.67]	-0.12 [-2.19]	0.04 [0.69]	0.10 [2.06]	0.25 [4.95]	0.34 [4.33]
α_{FF3}	-0.08 [-1.47]	-0.07 [-1.27]	0.09 [1.83]	0.10 [2.01]	0.24 [4.63]	0.32 [3.98]
α_{FF4}	-0.07 [-1.33]	-0.04 [-0.69]	0.09 [1.81]	0.06 [1.25]	0.21 [4.02]	0.28 [3.49]
α_{FF5}	-0.11 [-1.92]	0.01 [0.21]	0.11 [2.09]	0.01 [0.27]	0.14 [2.73]	0.24 [3.02]
α_{FF6}	-0.10 [-1.78]	0.03 [0.57]	0.11 [2.08]	-0.01 [-0.20]	0.12 [2.43]	0.22 [2.74]
Panel B: Fama and French (2018) 6-factor model loadings for EDP-sorted portfolios						
β_{MKT}	0.94 [69.20]	1.01 [79.17]	0.99 [78.05]	0.98 [83.03]	0.96 [79.79]	0.03 [1.49]
β_{SMB}	-0.03 [-1.52]	0.04 [2.27]	0.09 [5.01]	-0.05 [-2.98]	0.01 [0.54]	0.04 [1.38]
β_{HML}	-0.00 [-0.13]	-0.13 [-5.41]	-0.20 [-8.28]	-0.07 [-3.23]	-0.07 [-3.12]	-0.07 [-1.85]
β_{RMW}	0.10 [3.76]	-0.15 [-6.15]	-0.07 [-2.68]	0.10 [4.31]	0.13 [5.62]	0.03 [0.88]
β_{CMA}	-0.04 [-1.06]	-0.11 [-3.05]	0.05 [1.37]	0.25 [7.54]	0.29 [8.33]	0.33 [5.90]
β_{UMD}	-0.01 [-0.69]	-0.03 [-2.15]	-0.00 [-0.10]	0.03 [2.81]	0.02 [1.59]	0.03 [1.46]
Panel C: Average number of firms (n) and market capitalization (me)						
n	578	617	528	579	629	
me (\$10 ⁶)	1487	1171	1740	1882	2203	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [4.25]	0.34 [4.33]	0.32 [3.98]	0.28 [3.49]	0.24 [3.02]	0.22 [2.74]
Quintile	NYSE	EW	0.56 [7.97]	0.63 [9.02]	0.55 [8.30]	0.47 [7.16]	0.40 [6.35]	0.35 [5.53]
Quintile	Name	VW	0.33 [4.20]	0.34 [4.33]	0.31 [3.99]	0.29 [3.66]	0.25 [3.16]	0.24 [3.00]
Quintile	Cap	VW	0.23 [2.87]	0.23 [2.82]	0.22 [2.68]	0.18 [2.16]	0.18 [2.21]	0.15 [1.85]
Decile	NYSE	VW	0.33 [3.43]	0.29 [3.02]	0.27 [2.75]	0.21 [2.14]	0.22 [2.24]	0.18 [1.81]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [3.79]	0.30 [3.85]	0.28 [3.49]	0.26 [3.24]	0.22 [2.79]	0.21 [2.64]
Quintile	NYSE	EW	0.36 [4.73]	0.45 [5.89]	0.37 [5.14]	0.33 [4.63]	0.22 [3.33]	0.20 [3.02]
Quintile	Name	VW	0.29 [3.73]	0.30 [3.82]	0.27 [3.48]	0.26 [3.31]	0.22 [2.86]	0.22 [2.78]
Quintile	Cap	VW	0.19 [2.43]	0.19 [2.37]	0.18 [2.21]	0.16 [1.93]	0.15 [1.92]	0.14 [1.73]
Decile	NYSE	VW	0.29 [3.02]	0.25 [2.57]	0.23 [2.33]	0.19 [2.01]	0.19 [1.93]	0.17 [1.70]

Table 3: Conditional sort on size and EDP

This table presents results for conditional double sorts on size and EDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDP and short stocks with low EDP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDP Quintiles					EDP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.36 [1.36]	0.70 [2.60]	0.91 [3.34]	0.95 [3.79]	0.95 [4.05]	0.61 [6.39]	0.67 [7.11]	0.60 [6.68]	0.55 [6.08]	0.45 [5.11]	0.42 [4.72]
	(2)	0.58 [2.46]	0.69 [2.82]	0.84 [3.43]	0.96 [4.12]	0.90 [4.09]	0.32 [3.63]	0.38 [4.36]	0.28 [3.45]	0.25 [2.96]	0.20 [2.45]	0.18 [2.15]
	(3)	0.56 [2.78]	0.63 [2.80]	0.77 [3.38]	0.82 [3.92]	0.92 [4.63]	0.36 [4.26]	0.37 [4.39]	0.31 [3.71]	0.30 [3.47]	0.26 [3.08]	0.26 [2.95]
	(4)	0.49 [2.68]	0.63 [3.08]	0.78 [3.76]	0.82 [4.27]	0.78 [4.29]	0.29 [3.54]	0.30 [3.63]	0.23 [2.90]	0.24 [2.97]	0.05 [0.68]	0.08 [1.01]
	(5)	0.51 [3.29]	0.47 [2.61]	0.53 [2.98]	0.54 [3.26]	0.70 [4.50]	0.19 [1.99]	0.17 [1.79]	0.17 [1.74]	0.14 [1.39]	0.14 [1.45]	0.12 [1.22]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDP Quintiles					EDP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	330	331	331	328	324	24	27	33	25	23	
	(2)	92	90	91	91	89	47	46	47	47	46	
	(3)	66	65	65	65	65	81	78	80	81	83	
	(4)	56	56	56	56	56	176	174	182	181	185	
(5)	51	51	51	51	51	1274	1146	1373	1408	1619		

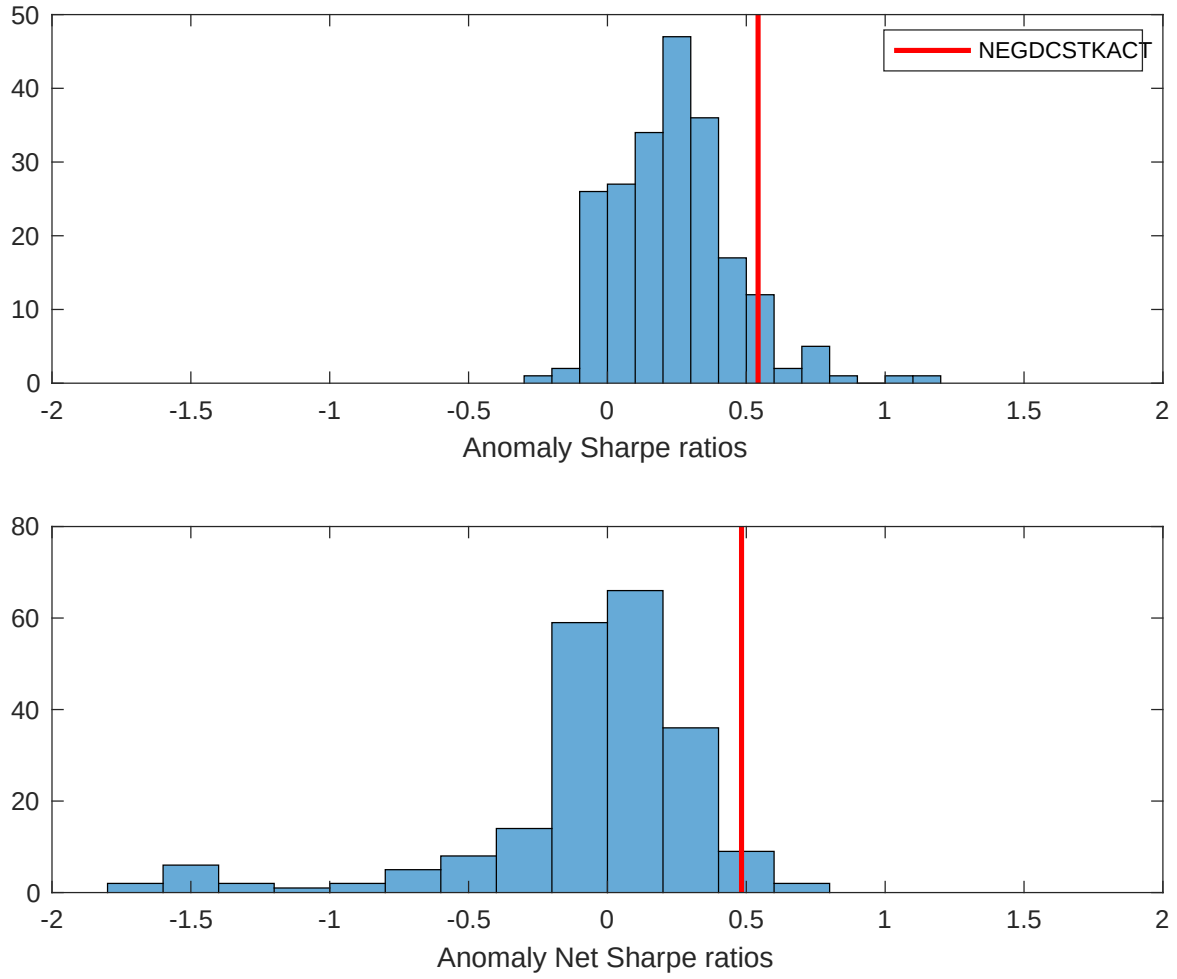


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

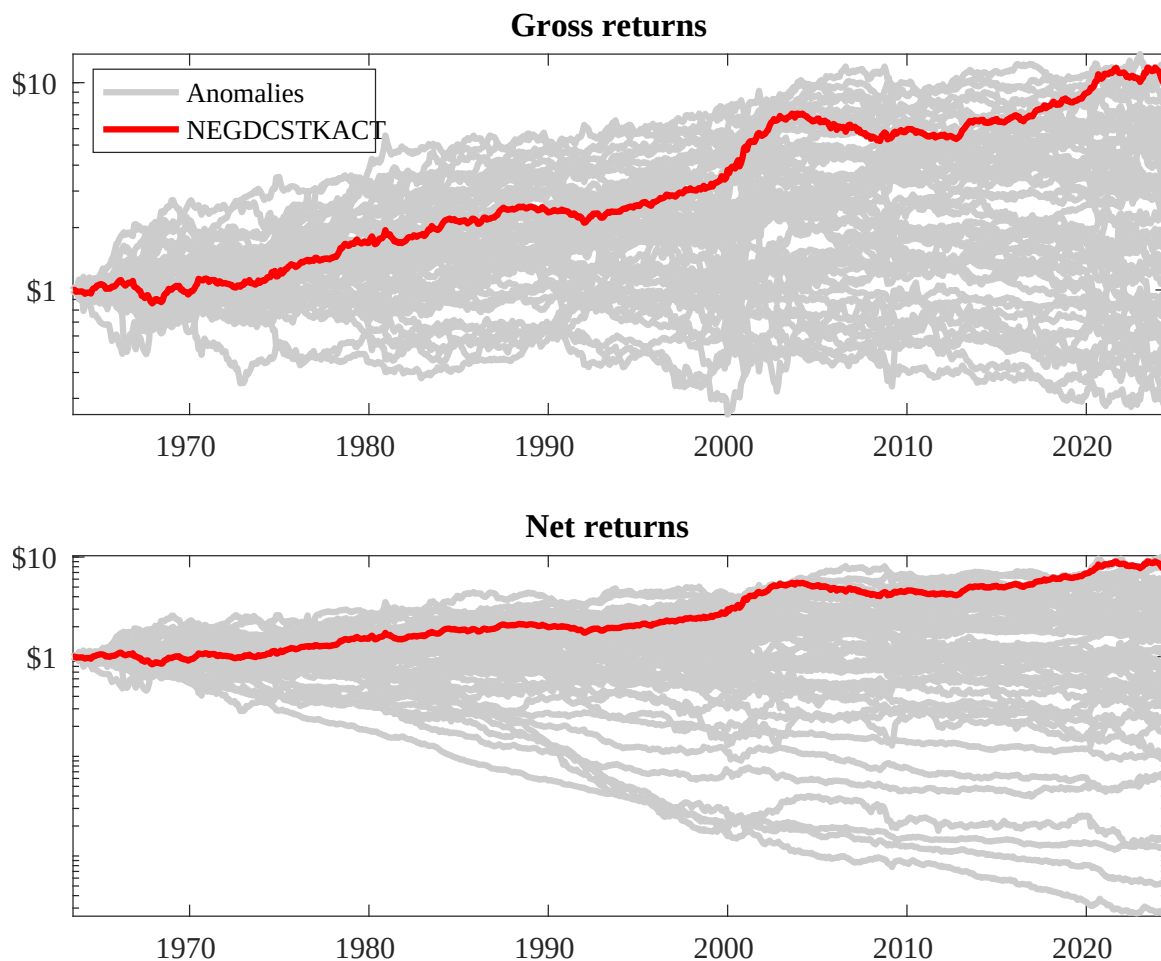
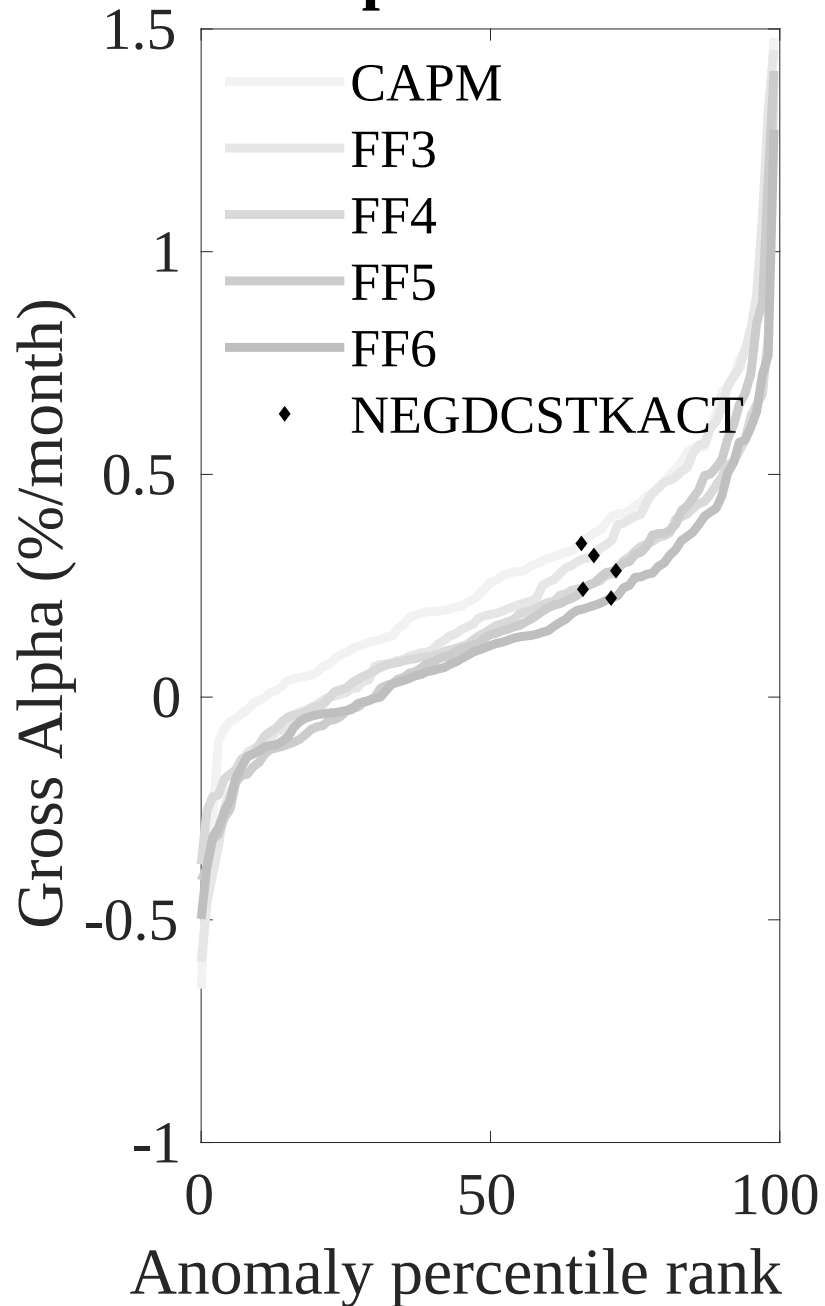


Figure 3: Dollar invested.

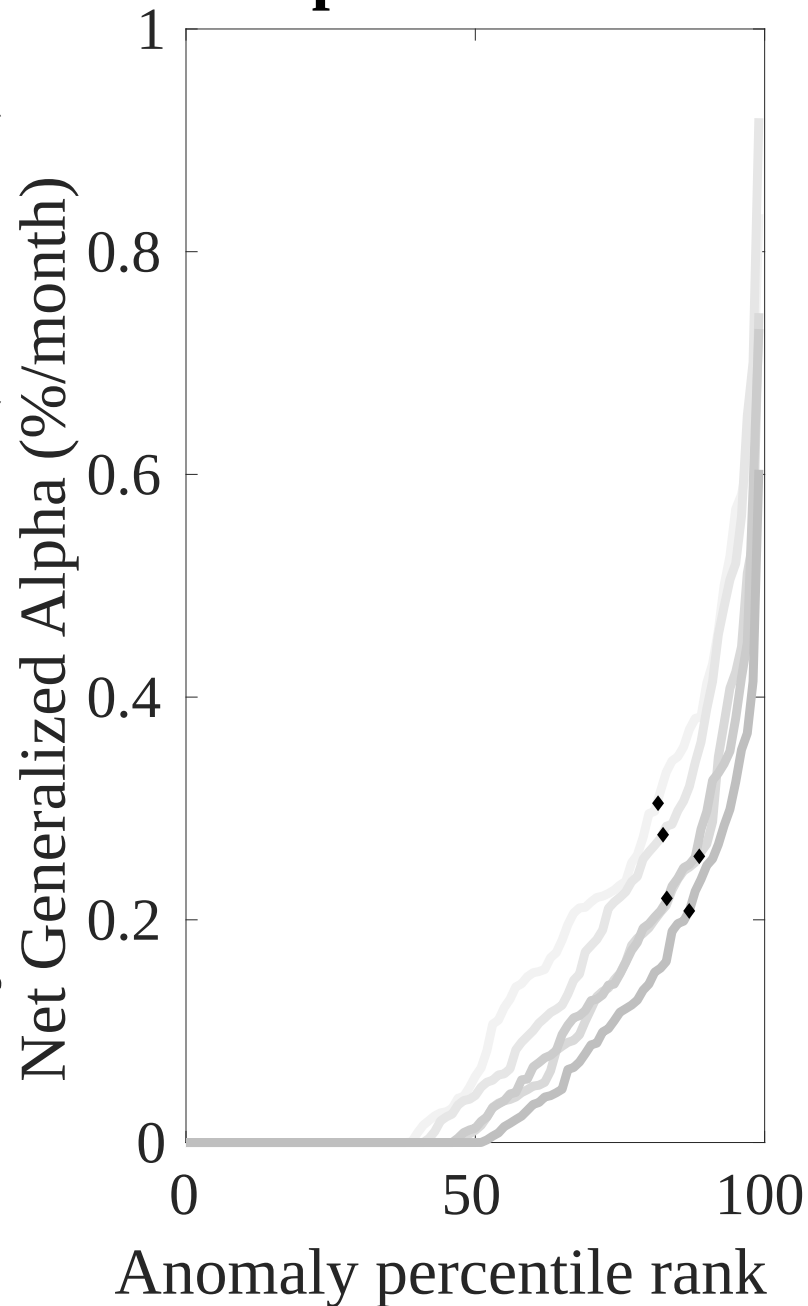
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



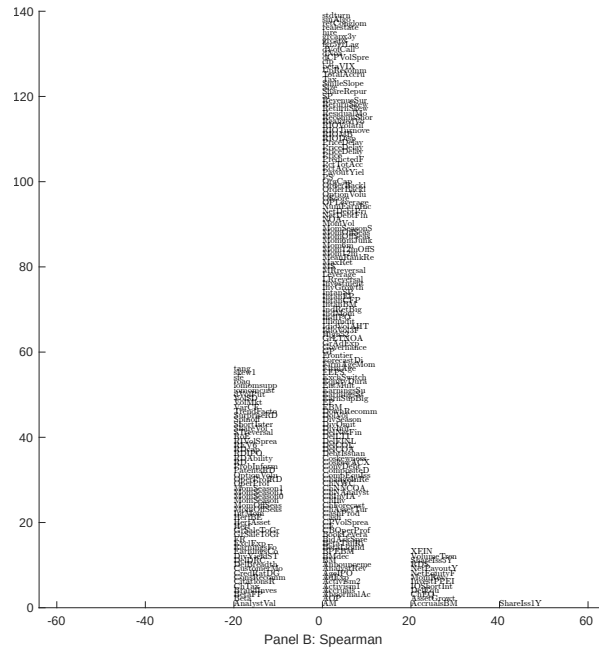
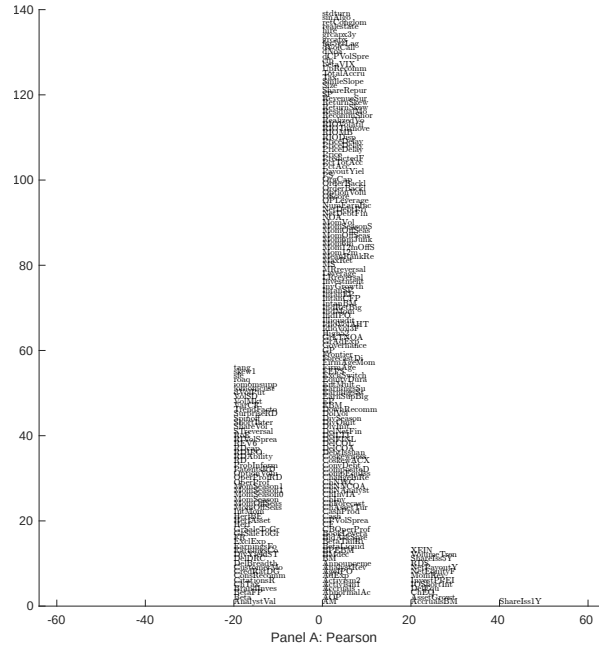


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with EDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

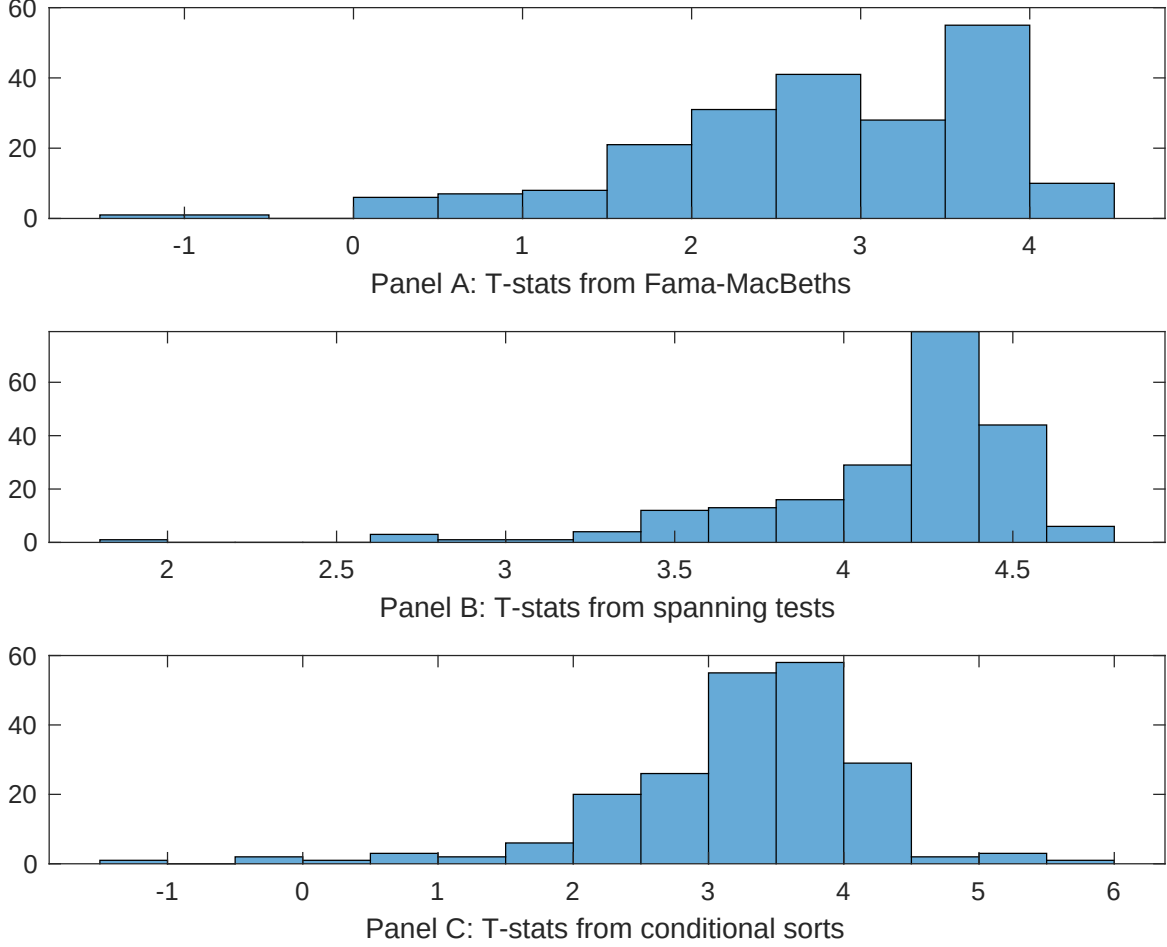


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.89]	0.13 [6.30]	0.17 [6.71]	0.12 [5.39]	0.13 [5.51]	0.13 [5.82]	0.11 [3.68]
EDP	0.14 [2.42]	0.12 [2.46]	0.14 [2.11]	0.12 [2.12]	0.18 [2.54]	0.15 [2.24]	0.77 [1.47]
Anomaly 1	0.14 [5.85]						0.59 [2.16]
Anomaly 2		0.27 [4.87]					0.25 [4.33]
Anomaly 3			0.39 [3.21]				-0.18 [-0.75]
Anomaly 4				0.21 [1.84]			0.10 [0.98]
Anomaly 5					0.12 [3.28]		0.30 [0.48]
Anomaly 6						0.13 [9.32]	0.83 [5.96]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.56]	0.15 [1.85]	0.20 [2.61]	0.25 [3.09]	0.22 [2.73]	0.20 [2.50]	0.16 [2.11]
Anomaly 1	29.42 [7.42]						13.47 [2.79]
Anomaly 2		28.62 [6.81]					14.35 [2.99]
Anomaly 3			31.52 [7.45]				30.12 [4.96]
Anomaly 4				16.03 [5.19]			4.67 [1.38]
Anomaly 5					14.56 [3.65]		-11.13 [-1.98]
Anomaly 6						9.80 [2.16]	3.36 [0.76]
mkt	3.76 [2.01]	4.32 [2.30]	4.08 [2.19]	5.53 [2.83]	2.78 [1.45]	3.10 [1.61]	5.94 [3.14]
smb	6.24 [2.31]	6.98 [2.56]	4.01 [1.49]	7.66 [2.74]	4.88 [1.76]	5.17 [1.86]	6.57 [2.43]
hml	-6.53 [-1.81]	-3.44 [-0.95]	-9.61 [-2.64]	-11.93 [-3.07]	-7.65 [-2.05]	-6.47 [-1.73]	-9.37 [-2.49]
rmw	-12.48 [-2.95]	-6.80 [-1.72]	4.95 [1.36]	-5.87 [-1.42]	4.79 [1.28]	3.53 [0.94]	-11.20 [-2.50]
cma	13.62 [2.34]	18.00 [3.17]	1.33 [0.20]	18.35 [3.08]	17.21 [2.55]	24.55 [3.82]	-7.90 [-1.09]
umd	1.59 [0.86]	1.07 [0.57]	2.39 [1.29]	4.22 [2.23]	3.30 [1.74]	2.76 [1.44]	0.86 [0.46]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	12	11	12	9	7	6	17

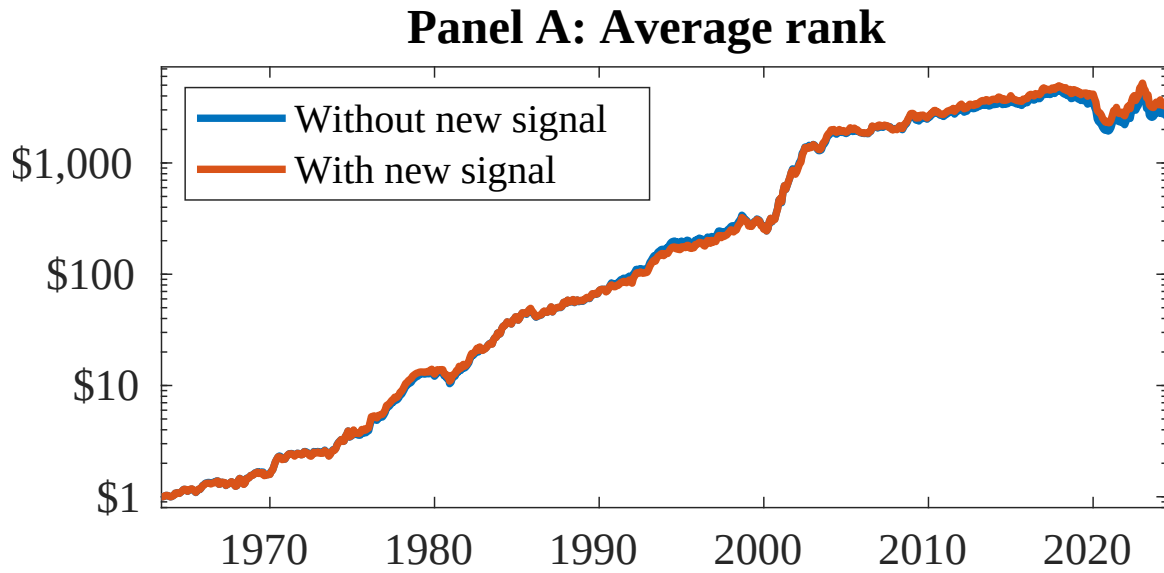


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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